KUSH PATEL

AML / Financial Crime Analyst | Transaction Monitoring · KYC / CDD · Investigations · FINTRAC Reporting | SQL · Python · Power BI

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Open to remote and relocation across Canada · Legally eligible to work in Canada · Available immediately

SUMMARY

- Analyst with 4+ years in finance and controls: month-end GL/P&L reconciliations and risk reporting at an investment firm, then reconciliation controls, exception logging and access controls as sole analyst at a distributor, cutting recurring discrepancies an estimated ~45%.
- Built a transaction-monitoring system on synthetic data that finds all 60 planted suspicious cases where a flat $10,000 rule finds 38, while alerting on a third fewer entities, with each rule mapped to FINTRAC's indicator categories and reporting obligations.
- Writes for the reviewer: an analytic brief on a structuring case with a bottom line, key judgments with confidence levels, alternatives considered, a reporting position and the facts still needed before a decision.

SKILLS

Financial Crime: KYC/CDD (customer risk rating, EDD triggers, PEP handling, beneficial ownership at 25% direct and indirect), sanctions name screening (fuzzy matching, transliteration, threshold tuning), structuring and smurfing typologies, rule design and tuning, alert triage and prioritisation, investigation case files, analytic briefs, PCMLTFA and FINTRAC reporting (LCTR including the 24-hour rule, EFTR, suspicious transaction report timing), threshold versus suspicion

Detection Analytics: SQL window functions for velocity and aggregation rules, entity-relative z-scores, anomaly detection (IsolationForest, Mahalanobis distance, gradient boosting), precision and recall by typology, cost curves for alert volume

Controls & Audit: reconciliation and control totals, journal-entry testing, segregation of duties, exception logs, evidence integrity, RLS/RBAC and column masking

Tools: SQL (T-SQL, SQLite), Python (pandas, scikit-learn), Power BI, Excel, Streamlit, Tableau, Git; Kafka streaming

PROFESSIONAL EXPERIENCE

Two Rivers Specialty Meats

Dec 2023 – Jun 2026 · North Vancouver, BC

Data Analyst — Operations, Logistics & Enterprise Reporting

- Built reconciliation controls comparing source against curated totals for GL/P&L and inventory valuation, cutting discrepancies an estimated ~45% and reporting errors ~30% by resolving them at the feed (internal figures, directional rather than audited).
- Automated data-quality tests (schema, completeness, uniqueness, referential integrity) with anomaly thresholds, exception logging and pipeline alerting.
- Delivered AP and AR aging, invoice-match exceptions and credit-exposure reporting for Finance.
- Implemented FEFO and lot-level traceability for recall readiness.
- Secured self-service reporting as access widened with row-level security, RBAC and column masking.
- As sole analyst, integrated ERP, Square POS, workforce and REST API feeds into one curated data mart in Microsoft Fabric and Azure Data Factory.

Shivam Investments

Dec 2020 – Jul 2022 · Vadodara, India (hybrid)

Financial Analyst — Data & Reporting

- Performed monthly GL/P&L reconciliations across sources and investigated discrepancies to keep financial reporting accurate and traceable.
- Delivered Power BI dashboards for portfolio and risk stakeholders with auditable measure definitions, and automated recurring analysis with Python and SQL, cutting manual effort an estimated ~40%.

SELECTED PROJECTS — public code; all entities and transactions are synthetic

Transaction Monitoring — SQL, Python, scikit-learn, SQLite, Streamlit · synthetic data

github.com/KushPatel29/aml-transaction-monitoring · live: aml-transaction-monitoring.streamlit.app

Layered detection over 100,299 transactions and 1,500 entities: five explainable rules plus an anomaly model find 60 of 60 planted cases where a flat $10,000 rule finds 38, alerting on 366 entities instead of 555. The alerts become 366 controlled investigations, 78 of them P0 with a four-hour review clock, worked against five typology hypotheses that each name a lawful lookalike and the evidence that tells them apart. Scores prioritise evidence; they never decide suspicion, and the record shows zero dispositions and zero filings.

FINTRAC mapping and an analytic brief — PCMLTFA, LCTR, EFTR, STR · synthetic case

regulatory mapping · analytic brief INV-ENT-01216

Threshold reports versus suspicion: each rule mapped to FINTRAC's indicator categories and the report it relates to, including why a $10,000 threshold rule cannot see structuring. The brief works one case: six cash deposits on six consecutive days, each just under $10,000, totalling $53,374.51 in 4.9 days, nine times the subject's usual weekly cash; assessed as likely sized to avoid the threshold (moderate confidence), with the three facts needed before a reporting decision.

Sanctions screening on Canada's real list — Python, rapidfuzz · real public list, synthetic customers

github.com/KushPatel29/aml-transaction-monitoring

Name screening that is measured, not assumed: customer names screened against the Consolidated Canadian Autonomous Sanctions List (5,708 entries, 9,604 names with aliases, Cyrillic transliterated). The alert threshold is set on one half of the list by a rule fixed in advance and reported on the other: in October 2026 it caught 97.2% of the name variants customers type, at 10 alerts per 10,000 synthetic customers. The list is downloaded at run time and never stored.

KYC: customer risk rating and beneficial ownership — Python, pandas · synthetic customers

github.com/KushPatel29/aml-transaction-monitoring

Who stands behind the account: 1,500 customers rated at onboarding and at a yearly refresh on stated factors, each rating with its reasons; a foreign PEP is high risk whatever the score, and every high-risk client is flagged for enhanced due diligence. For 450 businesses, beneficial owners traced through holding companies by multiplying shares down every chain to the 25% line, with opaque jurisdictions, loops and no-25%-owner cases flagged. Only 11 of the 60 planted-case customers rated high: a rating decides scrutiny, not suspicion.

GL / P&L Reconciliation — T-SQL, SQLite, Power BI, Tableau · synthetic data

github.com/KushPatel29/gl-reconciliation-dashboard

Controls that refuse to fake a sign-off: a close-certification engine tests all 5,400 journal entries and enforces segregation of duties; at month-end 8 of 60 account-periods certify, 9 go to a named reviewer and 43 are blocked.

Privacy assessment (project): PIPEDA-structured data-flow and risk assessment for the synthetic KYC workflow: 105 columns across 14 files classified in a machine-readable inventory, masked case outputs, and tested inventory coverage. Organizational controls still required before real-client use are documented.

EDUCATION

Northeastern University — Master of Professional Studies in Analytics (Applied Machine Intelligence), GPA 3.76/4.0 · Vancouver, BC — **Sep 2022 – Jun 2024**

Gujarat Technological University — Bachelor of Engineering, Computer Science · India — **Aug 2017 – Aug 2021**

CERTIFICATIONS

Google Data Analytics Professional Certificate · UC Davis SQL for Data Science · IBM Python for Data Science, AI & Development · Power BI Data Modelling with DAX · Databricks Lakehouse Fundamentals.